Circuit Debugging Assistant
Circuit Diagnostic Report

Report ID: ANALYSIS-4D9E62BC
Date: 2026-10-14 00:12
Circuit: Rlc Band Pass

STATUS:
POTENTIAL ISSUES IDENTIFIED

1. Uploaded versus Expected Circuit

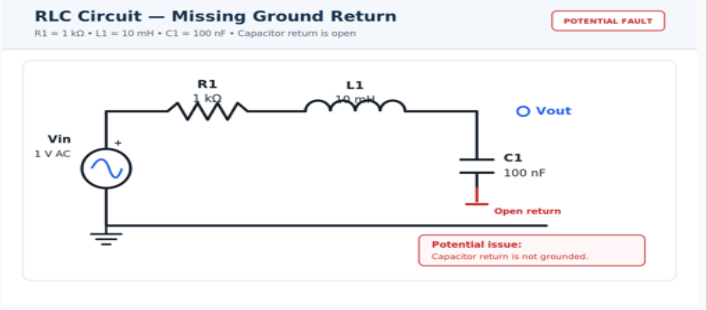


Figure 1A: Uploaded Circuit Schematic

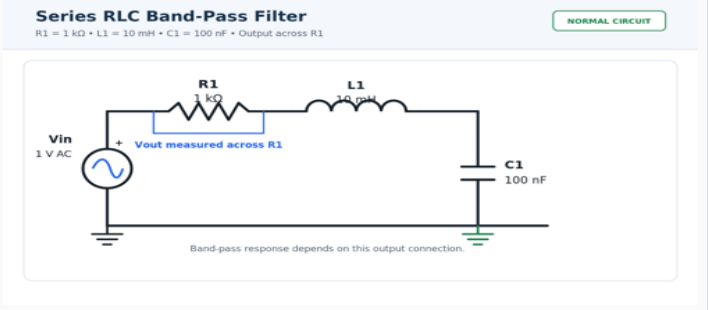


Figure 1B: Expected / Reference Circuit Model

2. Diagnosis Summary

The device under test was analyzed as a **Rlc Band Pass**. Numerical multi-evidence evaluation identified a probable parameter shift on **C -20%** (normalized residual score: **10.2864**). A total of 1 structural or parameter observations were cataloged.

3. Detected Flaws & Consistency Register

Severity	Component / Net	Finding & Evidence	Recommended Action
Major	**Net 'in'**	Node 'in' is connected to only 1 component terminal: ['R'].	Check for missing wire or unconnected terminal on net 'in'.
Major	**C**	Normalized residual S(f)=10.286 matches C -20% signature.	Verify C value with bench meter or replace with nominal part.

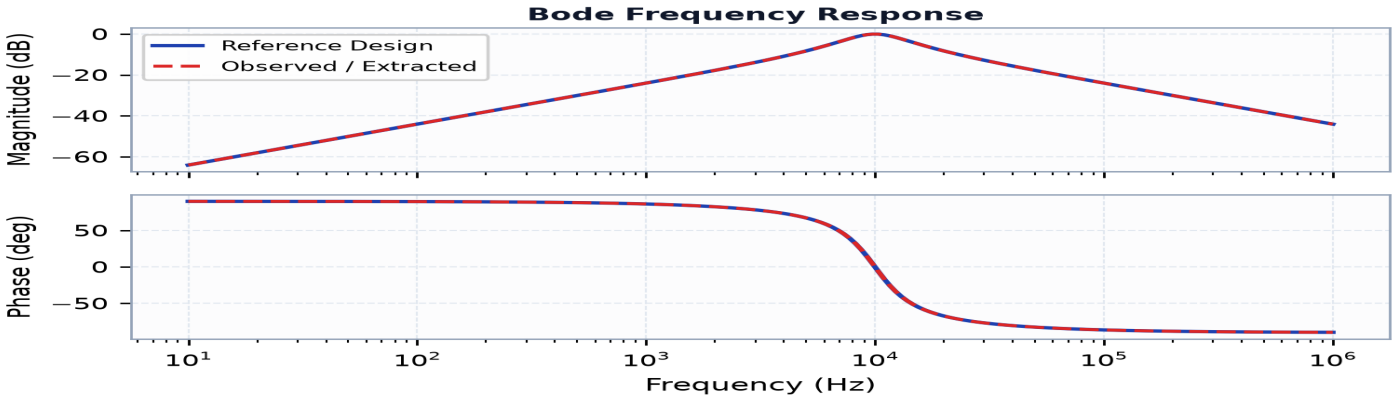
4. Confirmed Component Values

Reference	Component Type	Extracted / Nominal Value	Tolerance	Verification Status
R	Film Resistor	1000 Ω	±5.0%	Confirmed
L	Inductor Choke	10.00 mH	±10.0%	Confirmed
C	Ceramic / Electrolytic Capacitor	25.33 nF	±10.0%	Confirmed

5. Expected Behaviour & Governing Calculations

- **Resonant Center Frequency:** f0 = 1 / (2π √(L · C)) = **10000.06 Hz**
- **Quality Factor & Bandwidth:** Q = **1.59**, BW = **6283.26 Hz**
- **DC Bias Current:** I_DC = Vin / R = **1.00 mA**

6. Numerical Simulation Curves



7. Recommended Engineering Actions

1. **Physical Verification:** Perform **Final DC Current through R1** to independently separate ambiguous parameter hypotheses.
2. **Signal Integrity:** Measure test points with a 10:1 passive probe to avoid probe capacitance loading.
3. **Post-Repair Gate:** Confirm that measured response stays within ±5% of reference cutoff before releasing design.